WoSM Academy

MAT231-F27 · CALCULUS II

Gradient descent

The algorithm, and its convergence for convex quadratics

Dr. Ali Khudiyev
WoSM Academy

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Last time

- Critical points: $\nabla f = \mathbf{0}$. Local extrema in the interior are critical points.
- Second-derivative test: the signs of the eigenvalues of the Hessian.
- Convex functions: every critical point is a global minimum.
- Least squares is convex; its minimisers solve the normal equations.

But solving $\nabla f = \mathbf{0}$ is usually the hard part. Today: an algorithm that does not solve it, but walks towards a solution.

This lecture

01 The algorithm

02 Convex quadratics

03 Beyond quadratics

04 Summary and practice

Reading: Boyd & Vandenberghe, *Convex Optimization*, Sections 9.1–9.3.

01

The algorithm

Why iterate?

- $\nabla f(\mathbf{x}) = \mathbf{0}$ is a system of n equations in n unknowns. If f is not quadratic, the system is nonlinear, and usually has no formula for its solutions.
- Even for least squares, solving the $n \times n$ normal equations by elimination takes about n^3 operations: too many when n is in the millions.
- A gradient is cheap: by backpropagation (Lecture 12) it costs a small multiple of one evaluation of f.

> **Idea**
>
> Start somewhere, and repeatedly take a small step downhill. The steepest way down is $-\nabla f$ (Lecture 11).

Gradient descent

Algorithm

Choose a starting point $\mathbf{x}_0$, a **step size** $\eta > 0$ and a tolerance $\varepsilon > 0$. For $k = 0, 1, 2, \ldots$

1. compute $\mathbf{g}_k = \nabla f(\mathbf{x}_k)$;
2. if $\|\mathbf{g}_k\| \le \varepsilon$, stop;
3. set $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta\, \mathbf{g}_k$.

- In machine learning η is called the **learning rate**.
- The stopping rule asks for an approximate critical point. For a convex f that is close to a global minimum (made precise later today).
- The questions: does $\mathbf{x}_k$ converge, to what, how fast — and how to choose η?

Each step goes downhill — if it is short enough

By the first-order Taylor formula, with $\mathbf{g} = \nabla f(\mathbf{x}) \neq \mathbf{0}$:

$$f(\mathbf{x} - \eta\mathbf{g}) = f(\mathbf{x}) - \eta\,\|\mathbf{g}\|^2 + o(\eta) \;<\; f(\mathbf{x}) \qquad \text{for small } \eta > 0.$$

Example ($f(x) = x^2$ in one variable)

$x_{k+1} = x_k - 2\eta\, x_k = (1 - 2\eta)\, x_k$, so $x_k = (1 - 2\eta)^k x_0$.

- $0 < \eta < \frac{1}{2}$: steady approach to 0. $\eta = \frac{1}{2}$: 0 in one step.
- $\frac{1}{2} < \eta < 1$: the sign alternates, but $x_k \to 0$.
- $\eta \geq 1$: $|x_k|$ does not shrink: no convergence.

Too small a step is slow; too large a step overshoots. The rest of the lecture is this example in n dimensions.

02

Convex quadratics

The model problem

$$f(\mathbf{x}) = \tfrac{1}{2}\,\mathbf{x}^\top Q\mathbf{x} - \mathbf{b}^\top\mathbf{x}, \qquad Q \text{ symmetric positive definite,}$$

with eigenvalues $0 < m = \lambda_1 \le \cdots \le \lambda_n = L$. Then $\nabla f = Q\mathbf{x} - \mathbf{b}$ and $H_f = Q$.

- f is strictly convex, with the unique minimiser $\mathbf{x}^* = Q^{-1}\mathbf{b}$.

Why this model?

- Least squares is of this form: $\|A\mathbf{x} - \mathbf{b}\|^2 = \frac{1}{2}\mathbf{x}^\top(2A^\top A)\mathbf{x} - (2A^\top\mathbf{b})^\top\mathbf{x} + \|\mathbf{b}\|^2$.
- Near a strict local minimum $\mathbf{x}^*$, every C^2 function looks like this, with $Q = H_f(\mathbf{x}^*)$ (Taylor, Lecture 13).

The error recursion

Let $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$ be the error. Since $Q\mathbf{x}^* = \mathbf{b}$:

$$\mathbf{e}_{k+1} = \mathbf{x}_k - \eta\,(Q\mathbf{x}_k - \mathbf{b}) - \mathbf{x}^* = \mathbf{e}_k - \eta\, Q\mathbf{e}_k = (I - \eta Q)\,\mathbf{e}_k,$$

so $\mathbf{e}_k = (I - \eta Q)^k\,\mathbf{e}_0$.

Expand in an orthonormal basis of eigenvectors of Q (spectral theorem):

$$\mathbf{e}_0 = \sum_i c_i \mathbf{u}_i \quad\Longrightarrow\quad \mathbf{e}_k = \sum_i (1 - \eta\lambda_i)^k\, c_i\, \mathbf{u}_i.$$

> **In the eigenbasis, the problem splits**
>
> Each eigendirection is a one-variable problem like x^2: its error component is multiplied by $1 - \eta\lambda_i$ at every step.

Convergence

Theorem

Gradient descent on $f(\mathbf{x}) = \tfrac{1}{2}\mathbf{x}^{\top}Q\mathbf{x} - \mathbf{b}^{\top}\mathbf{x}$ converges to $\mathbf{x}^{*}$ from every starting point if and only if $0 < \eta < 2/L$. Then

$$\|\mathbf{x}_k - \mathbf{x}^{*}\| \le \rho^{k}\,\|\mathbf{x}_0 - \mathbf{x}^{*}\|, \qquad \rho = \max\bigl(|1-\eta m|,\ |1-\eta L|\bigr) < 1.$$

Proof.

$\|\mathbf{e}_k\|^2 = \sum_i (1-\eta\lambda_i)^{2k} c_i^2 \le \rho^{2k}\sum_i c_i^2 = \rho^{2k}\|\mathbf{e}_0\|^2$, because $|1-\eta\lambda_i| \le \rho$ for $m \le \lambda_i \le L$. And $\rho < 1$ exactly when $0 < \eta < 2/L$. Conversely, if $\eta \ge 2/L$ (or $\eta \le 0$), start at $\mathbf{e}_0 = \mathbf{u}_n$: then $\|\mathbf{e}_k\| = |1-\eta L|^{k} \ge 1$ for all k. □

The best step size

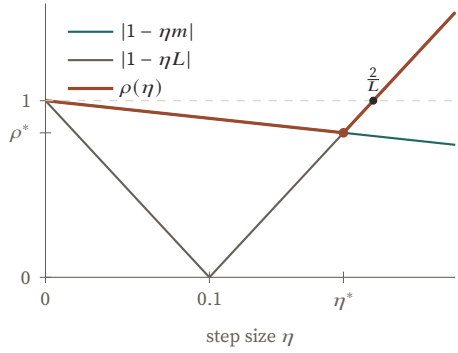
$\rho(\eta)$ is the larger of $|1-\eta m|$ (falling) and $|1-\eta L|$ (a V). It is smallest where they meet: $1-\eta m = \eta L - 1$.

Optimal fixed step

$$\eta^* = \frac{2}{m+L},$$

$$\rho^* = \frac{L-m}{L+m} = \frac{\kappa-1}{\kappa+1},$$

where $\kappa = L/m \geq 1$ is the **condition number** of Q.



$m = 1$, $L = 10$: $\eta^* = \frac{2}{11}$, $\rho^* = \frac{9}{11}$.

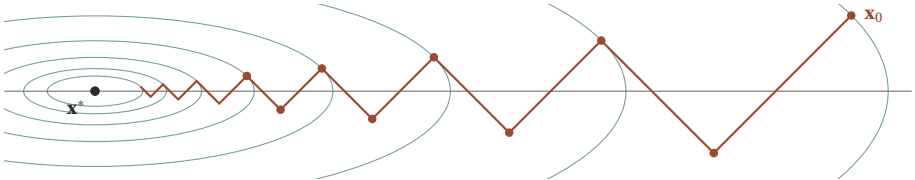
The condition number decides the speed

With $\eta = \eta^*$, the error falls by the factor 10^{-6} once $k \geq \dfrac{\ln 10^6}{\ln(1/\rho^*)}$:

κ	$\rho^* = \frac{\kappa-1}{\kappa+1}$	iterations
1	0	1
10	0.8182	69
100	0.9802	691
1000	0.9980	6908

- For large κ: $\ln(1/\rho^*) \approx 2/\kappa$, so about $\frac{\kappa}{2}\ln 10^6 \approx 6.9\,\kappa$ iterations: the work grows in proportion to κ.
- $\kappa = 1$ ($Q = mI$): the level sets are spheres, $-\nabla f$ points straight at $\mathbf{x}^*$, and one step of size $1/m$ lands on it.

Why it zigzags



$f = \frac{1}{2}(x^2 + 10y^2)$, $\kappa = 10$, $\eta = \eta^* = \frac{2}{11}$, from $\mathbf{x}_0 = (10, 1)$ (same scale on both axes).

- The steep direction (y, $\lambda = 10$) overshoots: factor $1 - \frac{20}{11} = -\frac{9}{11}$. The shallow direction (x, $\lambda = 1$) creeps: factor $\frac{9}{11}$.
- Each step is perpendicular to the level curve it starts from, not aimed at $\mathbf{x}^*$.

03

Beyond quadratics

The descent lemma

Lemma

Let f be C^2 on $\mathbb{R}^n$ with $\mathbf{v}^\top H_f(\mathbf{x})\mathbf{v} \le L\|\mathbf{v}\|^2$ for all $\mathbf{x}, \mathbf{v}$. Then

$$f(\mathbf{y}) \le f(\mathbf{x}) + \nabla f(\mathbf{x}) \cdot (\mathbf{y} - \mathbf{x}) + \tfrac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$

Proof. Let $\mathbf{h} = \mathbf{y} - \mathbf{x}$ and $\varphi(t) = f(\mathbf{x} + t\mathbf{h})$. By Taylor with the Lagrange remainder (Calculus I, Unit 17), $\varphi(1) = \varphi(0) + \varphi'(0) + \tfrac{1}{2}\varphi''(\tau)$ for some $\tau \in (0,1)$, and $\varphi''(\tau) = \mathbf{h}^\top H_f(\mathbf{x} + \tau\mathbf{h})\,\mathbf{h} \le L\|\mathbf{h}\|^2$. □

With $\mathbf{y} = \mathbf{x}_{k+1} = \mathbf{x}_k - \eta\mathbf{g}_k$:

$$f(\mathbf{x}_{k+1}) \le f(\mathbf{x}_k) - \eta\Big(1 - \frac{L\eta}{2}\Big)\|\mathbf{g}_k\|^2.$$

For $0 < \eta < 2/L$ every step decreases f. For $\eta = 1/L$: $f(\mathbf{x}_{k+1}) \le f(\mathbf{x}_k) - \frac{1}{2L}\|\mathbf{g}_k\|^2$.

Strongly convex functions

Suppose also $\mathbf{v}^\top H_f(\mathbf{x})\mathbf{v} \geq m\|\mathbf{v}\|^2$ with $m > 0$ (f is **strongly convex**). The same proof gives $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x}) \cdot (\mathbf{y} - \mathbf{x}) + \frac{m}{2}\|\mathbf{y} - \mathbf{x}\|^2$. Minimising the right-hand side over $\mathbf{y}$ (at $\mathbf{y} = \mathbf{x} - \nabla f(\mathbf{x})/m$):

$$f^* = \min f \;\geq\; f(\mathbf{x}) - \frac{1}{2m}\|\nabla f(\mathbf{x})\|^2.$$

A large gradient forces a large gap above the minimum, and the other way round: at a point with a small gradient, f is close to f^*.

Linear convergence

Theorem

If $m\|\mathbf{v}\|^2 \le \mathbf{v}^\top H_f(\mathbf{x})\mathbf{v} \le L\|\mathbf{v}\|^2$ for all $\mathbf{x}, \mathbf{v}$ (with $m > 0$), and $\eta = 1/L$, then $f(\mathbf{x}_k) - f^* \le \left(1 - \frac{1}{\kappa}\right)^k \big(f(\mathbf{x}_0) - f^*\big)$, where $\kappa = L/m$.

Proof. The descent lemma, then the bound $\|\mathbf{g}_k\|^2 \ge 2m\,\big(f(\mathbf{x}_k) - f^*\big)$:

$$f(\mathbf{x}_{k+1}) - f^* \le f(\mathbf{x}_k) - f^* - \frac{1}{2L}\|\mathbf{g}_k\|^2 \le \left(1 - \frac{m}{L}\right)\big(f(\mathbf{x}_k) - f^*\big). \qquad \square$$

“Linear”: the gap shrinks by a constant factor per step, so its logarithm falls linearly in k. Again the condition number sets the speed, as for quadratics (Boyd & Vandenberghe, Section 9.3).

Non-convex functions

If f is bounded below and the descent lemma holds, then with $\eta = 1/L$ the decreases add up:

$$\frac{1}{2L}\sum_{k=0}^{K}\|\mathbf{g}_k\|^2 \le f(\mathbf{x}_0) - f(\mathbf{x}_{K+1}) \le f(\mathbf{x}_0) - \inf f,$$

so $\nabla f(\mathbf{x}_k) \to \mathbf{0}$: gradient descent finds approximate **critical points.**

> **A critical point, not necessarily a minimum**
>
> $f = x^4 + y^4 - 4xy + 1$ (Lecture 14): from $(1, -1)$ the iterates stay on the line $y = -x$ and converge to the saddle $(0, 0)$. From $(1, -0.99)$ they converge to the minimum $(1, 1)$ ($\eta = 0.05$).

For convex f there is no such problem: every critical point is a global minimum.

Choosing the step size in practice

L is rarely known. Common choices:

- **Fixed** η, tuned by trial: too large diverges, too small crawls.
- **Backtracking line search**: start with $\eta = 1$ and halve it until

$$f(\mathbf{x}_k - \eta \mathbf{g}_k) \le f(\mathbf{x}_k) - \tfrac{\eta}{2}\|\mathbf{g}_k\|^2.$$

  By the descent lemma this holds as soon as $\eta \le 1/L$, so the search stops.
- **Exact line search**: minimise $f(\mathbf{x}_k - \eta \mathbf{g}_k)$ over η. For the model quadratic it has a formula: $\eta_k = \dfrac{\mathbf{g}_k^\top \mathbf{g}_k}{\mathbf{g}_k^\top Q\, \mathbf{g}_k}$.

All of these still zigzag when κ is large: the trouble is the direction, not the length of the step.

Beyond gradient descent

- **Newton's method:** $\mathbf{x}_{k+1} = \mathbf{x}_k - H_f(\mathbf{x}_k)^{-1}\nabla f(\mathbf{x}_k)$ jumps to the minimum of the quadratic Taylor model. On the model quadratic it reaches $\mathbf{x}^*$ in one step. Each step solves an $n \times n$ linear system (compare Calculus I, Unit 16).
- **Momentum** and **conjugate gradients** combine the new gradient with earlier steps to damp the zigzag.
- **Stochastic gradient descent**: when f is a sum over a huge data set, use the gradient of a randomly chosen part of the sum. With refinements, this is how neural networks are trained.

> **Project 2**
>
> Project 2 asks you to implement gradient descent and study how the step size affects convergence. Compare what you observe with the theory of this lecture.

04

Summary and practice

Summary

- **Gradient descent**: $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$; a short enough step always decreases f.
- **Convex quadratic** $\frac{1}{2}\mathbf{x}^\top Q\mathbf{x} - \mathbf{b}^\top \mathbf{x}$: $\mathbf{e}_{k+1} = (I - \eta Q)\mathbf{e}_k$. Convergence from every start iff $0 < \eta < 2/L$, at the rate $\rho = \max(|1 - \eta m|, |1 - \eta L|)$.
- Best fixed step $\eta^* = 2/(m + L)$, with $\rho^* = (\kappa - 1)/(\kappa + 1)$: the number of iterations grows like the condition number $\kappa = L/m$.
- Zigzag: steep directions overshoot while shallow ones creep.
- In general: the descent lemma; linear convergence for strongly convex f; for non-convex f, convergence to critical points, which may be saddles.

Practice

1. Take two steps of gradient descent on $f = x^2 + 2y^2$ from $(1, 1)$ with $\eta = 0.25$.
2. For $f = \tfrac{1}{2}\mathbf{x}^\top Q\mathbf{x}$ with $Q = \operatorname{diag}(1, 4)$: for which η does gradient descent converge? Find η^* and ρ^*.
3. The same for $Q = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.
4. With $\kappa = 50$ and $\eta = \eta^*$, how many iterations reduce the error by the factor 10^{-3}?
5. Show that one step of Newton's method minimises $\tfrac{1}{2}\mathbf{x}^\top Q\mathbf{x} - \mathbf{b}^\top\mathbf{x}$ from any $\mathbf{x}_0$.
6. Show that exact line search on the model quadratic gives $\eta_k = \mathbf{g}_k^\top\mathbf{g}_k / \mathbf{g}_k^\top Q\mathbf{g}_k$.
7. Gradient descent on $f(x) = x^4$ from $x_0 = 1$, with $\eta = 0.1$: show that $x_k \to 0$. Why does the convergence theorem for strongly convex functions not apply?

Answers

1. $\nabla f = (2x, 4y)$: $(1,1) \to (0.5, 0) \to (0.25, 0)$.
2. $0 < \eta < \tfrac{1}{2}$; $\eta^* = \tfrac{2}{5}$, $\rho^* = \tfrac{3}{5}$.
3. Eigenvalues 2 and 4: $0 < \eta < \tfrac{1}{2}$, $\eta^* = \tfrac{1}{3}$, $\rho^* = \tfrac{1}{3}$.
4. $\rho^* = \tfrac{49}{51}$; $k \geq \ln 1000 / \ln \tfrac{51}{49} \approx 172.7$: 173 iterations.
5. $\mathbf{x}_1 = \mathbf{x}_0 - Q^{-1}(Q\mathbf{x}_0 - \mathbf{b}) = Q^{-1}\mathbf{b} = \mathbf{x}^*$.
6. $\varphi(\eta) = f(\mathbf{x}_k - \eta \mathbf{g}_k) = f(\mathbf{x}_k) - \eta\, \mathbf{g}_k^\top \mathbf{g}_k + \tfrac{\eta^2}{2} \mathbf{g}_k^\top Q \mathbf{g}_k$; set $\varphi'(\eta) = 0$.
7. $x_{k+1} = x_k(1 - 0.4 x_k^2)$: positive and decreasing, and the limit c satisfies $c = c - 0.4c^3$, so $c = 0$. Slowly: $x_{100} \approx 0.109$. $f''(0) = 0$, so no $m > 0$ exists.

Next time

Lecture 16 — Constrained optimisation: Lagrange multipliers (Unit 16)

- Optimising $f(\mathbf{x})$ subject to a constraint $g(\mathbf{x}) = c$
- The condition $\nabla f = \lambda \nabla g$, and why it holds
- Examples from geometry, economics, information theory and linear algebra

Reading: Strang & Herman, *Calculus, Volume 3*, Section 4.8; Marsden & Tromba, *Vector Calculus*, Section 3.4.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.